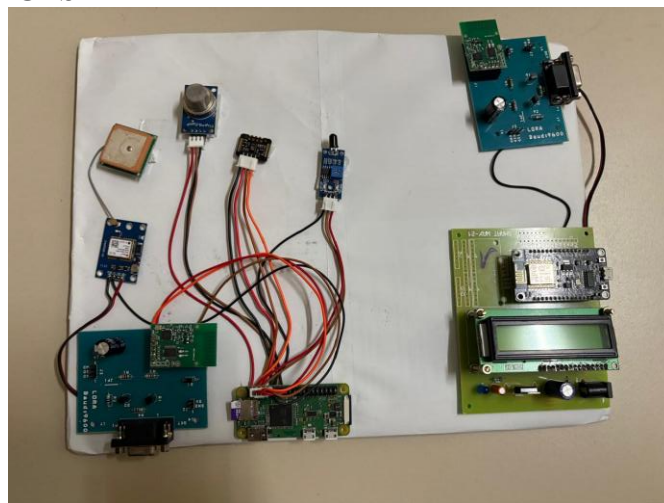
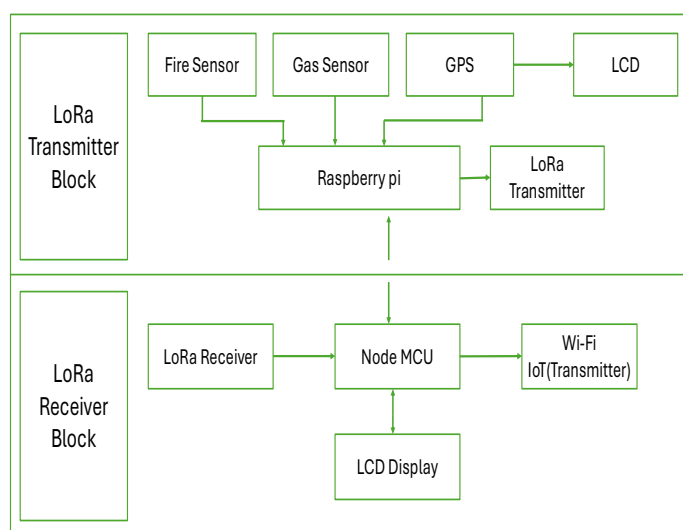


**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**LoRa- ENABLED SMART FAULT DETECTION AND MONITORING
PLATFORM FOR POWER DISTRIBUTION USING SELF POWERED
IOT DEVICES**



DESCRIPTION OF THE PROJECT: The LoRa Enabled IoT Power Distribution Monitoring System is designed to improve the safety and reliability of electrical power distribution networks. Electrical equipment such as transformers and distribution panels may face problems like overheating, gas leakage, or smoke generation due to faults. If these conditions are not detected early, they may lead to equipment damage, fire hazards, and power failures. The proposed system uses IoT technology with a Raspberry Pi controller to monitor environmental conditions around power distribution equipment. The system includes temperature, gas, and smoke sensors to continuously measure important parameters. These sensors detect abnormal conditions such as high temperature, harmful gases, or smoke generated due to electrical faults. The collected data is processed by the Raspberry Pi and transmitted through a LoRa communication module, which allows long-range wireless communication with low power consumption. When abnormal conditions are detected, the system sends an alert to the monitoring station. This system helps in early fault detection, improving safety and maintenance efficiency in power distribution systems.

REQUIREMENTS: Raspberry Pi, NodeMCU (ESP8266), LoRa Module (SX1278), GPS Module (NEO-6M), Temperature Sensor, Gas Sensor (MQ-2), Flame Sensor, Smoke Sensor, 16×2 LCD Display, Power Supply Circuit, GPS Antenna

PROJECT GUIDE:

STUDENTS MEMBERS: BATCH 2024-2028

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Outcome of the project: Nil

Collaboration with Industry: Nil